

Fig. 9: AttendanceRegistrationSystem



Fig. 10: New database name as EmployeeAttendanceDB

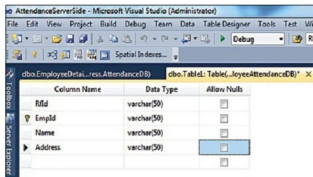


Fig. 11: EmployeeDetail table

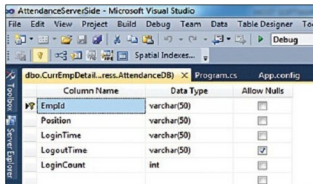


Fig. 12: CurrEmpDetail table

application AttendanceRegistrationSystem (refer Fig. 9)

2. On the left side of the window, there is Server Explorer tab (View -> Other windows -> Server Explorer).

Here, there are three nodes:

- (i) Data Connections
- (ii) Servers
- (iii) SharePoint Connections

Right-click on Data Connections and select Create New SQL Server Database. A new dialog box of this name opens.

3. Now, enter the server name as it appears when you open SQL Server Management Studio (already installed before). And, under Windows authentication mode, enter the new database name (let us say, EmployeeAttendanceDB) as shown in Fig. 10.

4. Add a new table by right-clicking on Tables and add following columns: Rfid, EmpId (right-click and set as primary key), Name and Address of type varchar(50). Save the table and name it EmployeeDetail. Uncheck Allow Nulls as shown in Fig. 11.

5. Similarly, add another table with following columns: EmpId (primary key), Position, LoginTime, LogoutTime and LoginCount. All variables are of type varchar(50) except LoginCount, which is int. Uncheck Allow Nulls for all except LogoutTime. Save the table as CurrEmpDetail as shown in Fig. 12.

By executing above steps, we have successfully created our database.

Designing server-side application—data-access layer

The data-access layer is on the server-side of our application, which is

used to access data from the database.

In data-access layer, we use ADO. Net Entity Framework for our development. C# language is object-based, which can be either Is-A or Has-A relationship. Our database is a relational model database, which means that each employee has various employee attributes, such as employee name, employee ID, employee RFID number, etc, and hence each employee has a (Has-A) number of attributes to its record, which together provide the complete information of the employee.

In order to follow this approach, we need to keep proper mapping between C# and the database table that we have created. This can be achieved by using object relational mapping (ORM). ORM is a tool or software-programming technique used in object-oriented programming for accessing/storing data from an object to a database.

While designing and building an application or service we divide these into three parts: domain model, logical model and physical model.

The domain model defines entities and relationships in the system. The logical model for a relational database normalises the entities and relationships into tables. Application programmers mostly confine themselves to the logical model by writing SQL queries and stored procedures. The physical model addresses capabilities of a particular data engine by specifying storage details such as partitioning and indexing—this part is mostly taken care of by database administrators to improve performance.

Following are the steps for designing the data-access layer:

1. Create data transferable object (DTO) and name it DataTransferObjectLib. Create a new project of type class library under AttendanceServer-Side solution.

2. Create two public classes, EmpDetailsModel and CurrDetailsModel, inside it as shown in Fig. 13.

Now, build this project by right-clicking on project DataTransferObjectLib.